

Lattice CT Inspection

Final Technical Submission

Part 1: Unit-cell segmentation and NDE
Part 2: Registered 0.5%-missing octet-lattice screening

Submission scope	Status
Part 1	Completed: segmentation, skeleton, and NDE visual report
Part 2	Completed: registered CT/graph candidate screening and sensitivity analysis
Quantitative confirmation	Not supported by available evidence; no missing struts confirmed

Prepared from the repository saved, traceable analysis artifacts. Unregistered STL files were excluded from quantitative CT comparisons.

Executive Summary

This submission consolidates the completed Part 1 and Part 2 outputs. Part 1 establishes a reproducible segmentation-to-NDE workflow on the unit-cell CT volume. Part 2 evaluates the supplied registered 0.5%-missing octet-lattice CT/JSON pair using a distance-tolerant centerline tube test and sensitivity checks.

Finding	Result	Interpretation
Part 1 selected threshold	0.005813	Otsu-selected balance of continuity and compactness.
Part 1 mask	717,852 voxels (4.2787%)	Selected mask for 256 x 256 x 256 volume.
Part 1 skeleton	3,182 voxels; 1 component	Connected structural proxy.
Part 2 design graph	18,468 struts; 10,206 junctions	Supplied registered JSON only.
Part 2 retained setting	Tube radius 2; threshold 39,725	Least flagging tested radius.
Part 2 confirmation	0 confirmed missing struts	Candidates are not physical defect labels.

1. Objectives and Data

The work has two objectives: (1) produce a traceable segmentation, skeletonization, and NDE workflow for the unit-cell volume; and (2) screen an aligned octet-lattice specimen for missing or disconnected strut candidates using only the supplied registered CT graph.

Dataset	Input	Use
Part 1 unit cell	data/unitcell/unitcell.npy	Segmentation and NDE.
Part 2 CT	0.5%-missing TIFF, 761 x 815 x 837 uint16	As-built image evidence.
Part 2 design	Registered JSON, 18,468 struts	Expected centerlines in CT coordinates.
STL variants	Unregistered	Excluded from quantitative comparison.

2. Part 1 Methods and Results

Three thresholds were compared. The selected threshold, 0.005813, is the Otsu value and provided a balanced foreground fraction with visually continuous lattice material. The chosen mask was skeletonized and rendered from two 3D perspectives for NDE review.

Threshold	Foreground fraction	Decision
0.004500	4.3416%	Lower candidate.
0.005813	4.2787%	Selected Otsu threshold.
0.007000	4.2480%	Higher candidate.

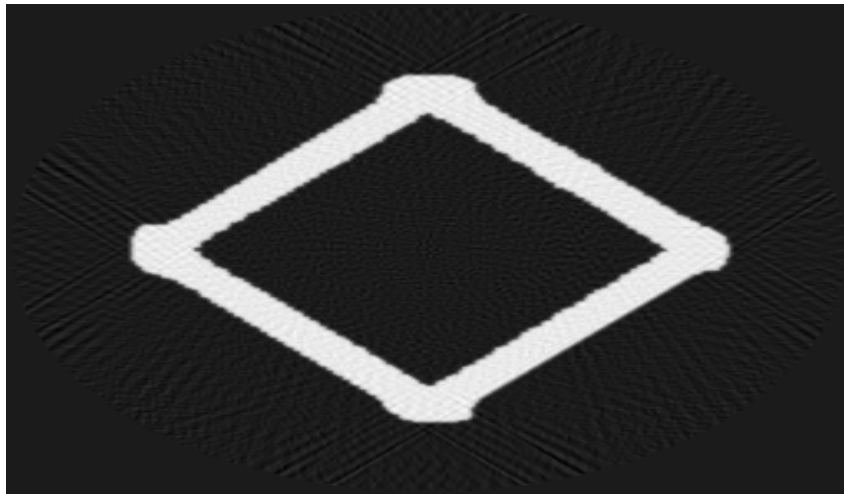


Figure 1. Part 1 unit-cell raw CT slice (slice 128).

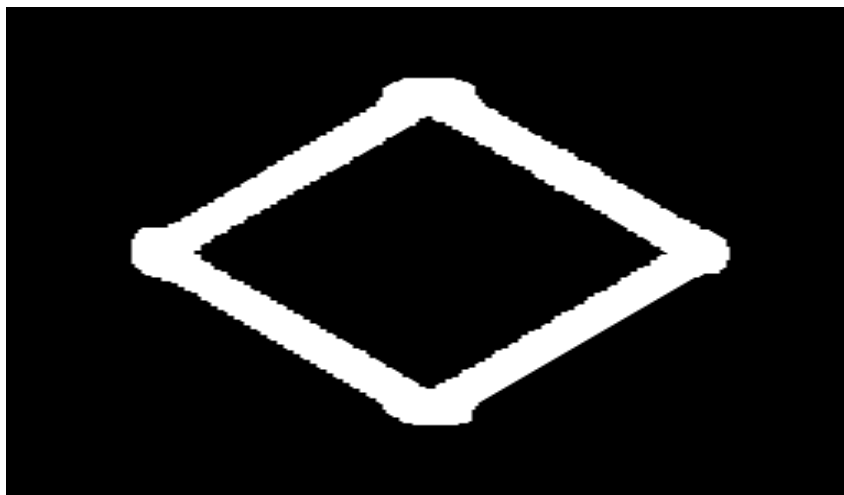


Figure 2. Part 1 selected segmentation mask at the same slice.

Part 1 NDE Visual Validation

3D Isosurface with Skeleton
(threshold=0.5)

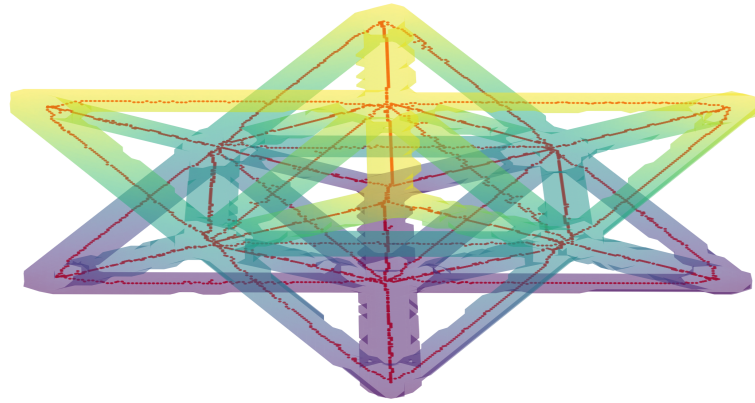


Figure 3. NDE View A: elevation 30 degrees, azimuth 45 degrees.

3D Isosurface with Skeleton
(threshold=0.5)

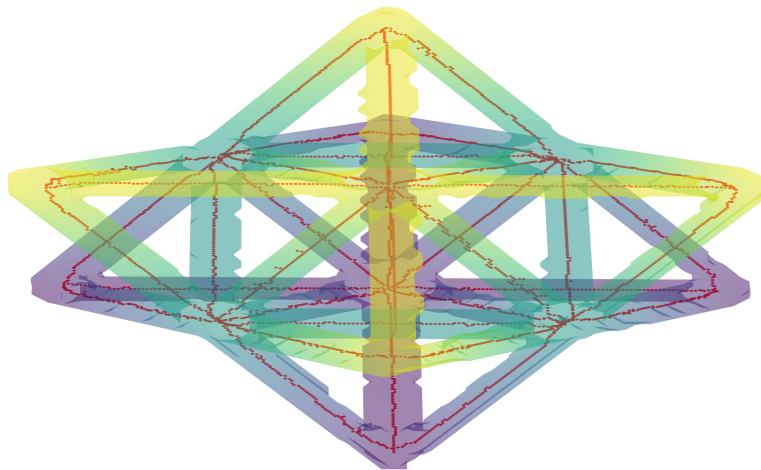


Figure 4. NDE View B: elevation 60 degrees, azimuth 45 degrees.

Part 1 conclusion: the selected mask and skeleton are dimensionally compatible with the source volume. The single connected skeletal component supports intended unit-cell connectivity; it is not a mechanical simulation.

3. Part 2 Registered CT-to-Graph Screening

The Part 2 graph was supplied in CT voxel coordinates. Expected strut centerlines were sampled after XYZ-to-CT indexing conversion. A local tube test used the 75th-percentile CT intensity around each sample, which is more tolerant of small offsets than a single centerline voxel. Samples were trimmed to the central 8%-92% of each strut to reduce junction influence.

Parameter	Retained value	Reason
Tube radius	2 CT voxels	Least flagging among radii 1, 2, and 3.
Threshold	39,725	Separation criterion retaining median support ≥ 0.60 .
Low support rule	Material fraction < 0.04	Conservative candidate screen.
Disconnected rule	Fraction < 0.35 with long gap	Separates gap-like behavior.

4. Part 2 Sensitivity and Candidate Results

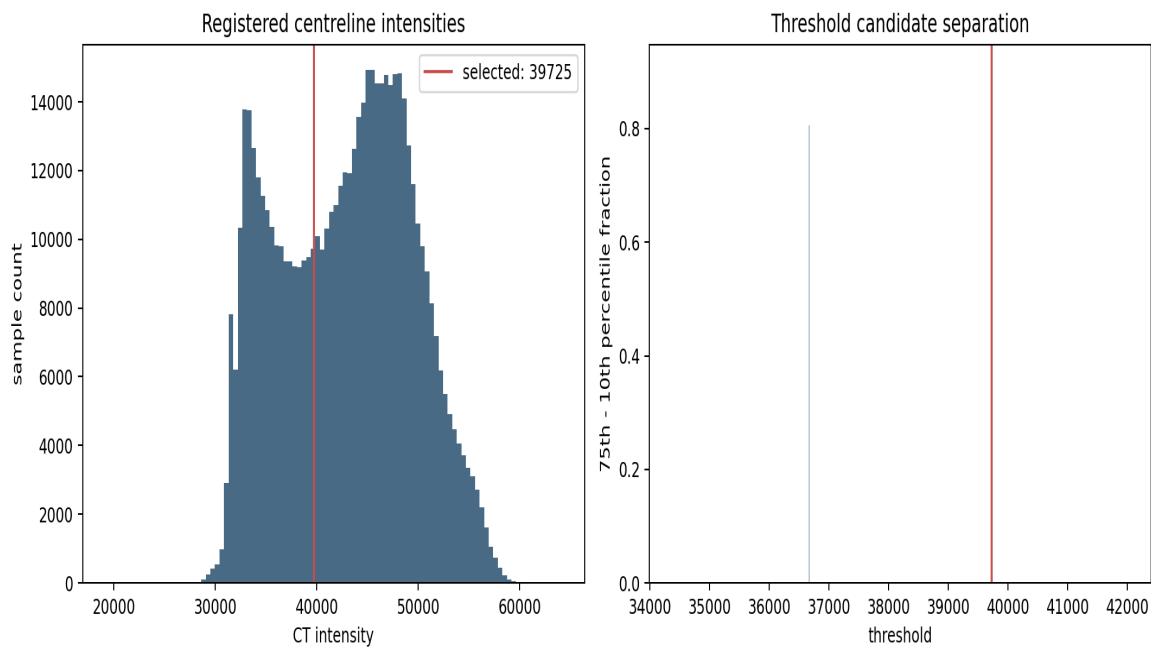


Figure 5. Radius-2 centerline intensity distribution and threshold-candidate separation.

Tube radius	Threshold	Low support	Disconnected	Uncertain	Clear
1	39,725	1,217	5,614	1,199	10,438
2 retained	39,725	1,049	5,262	1,083	11,074
3	42,017	1,159	6,505	1,384	9,420

At the retained radius-2 setting, 7,394 struts are review candidates: 1,049 low-support, 5,262 disconnected, and 1,083 uncertain. This is a screen, not a defect count. Dependence on tube radius and threshold is stronger evidence for residual registration, segmentation, and partial-volume effects than for thousands of physical defects.

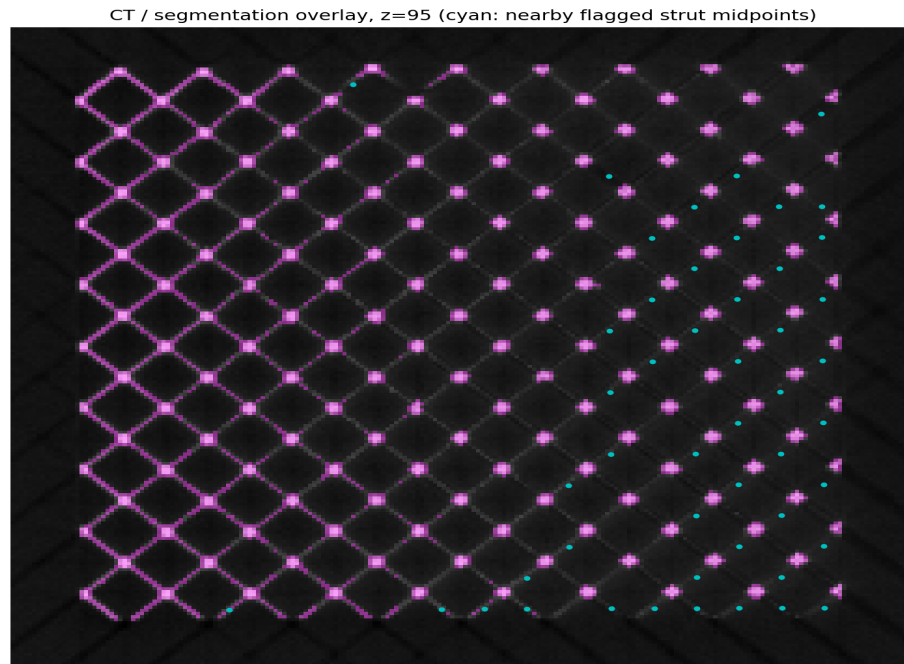


Figure 6. Supplemental CT/segmentation overlay. Supporting visual evidence only; not the authoritative result source.

5. Discussion, Limitations, and Recommendation

The nominal 0.5% label implies approximately 92 intentionally removed struts but was never used as a classification target. The screen returns far more candidates than the nominal label and does so inconsistently as parameters change. A valid report retains this discrepancy rather than tuning the result to meet the expectation.

The supplied JSON does not contain transform covariance, independent fiducials, or per-strut ground truth. Therefore an additional rigid or deformable registration refinement cannot be defensibly estimated from the available artifacts. The appropriate conclusion is zero confirmed physical missing struts, with a ranked candidate list for review.

Claim	Status	Evidence
Registered JSON and CT used	Supported	Paths and dimensions recorded in summaries.
Unregistered STL excluded	Supported	Workflow and provenance state this explicitly.
Candidates reproducible	Supported	CSV and diagnostics saved.
Physical defect count known	Not supported	No truth join or blinded review.
Mechanical repair benefit known	Not supported	Historical repair output is non-FEA only.

6. Recommended Next Steps

1. Obtain independent registration landmarks or transform uncertainty.
2. Perform blinded CT-neighborhood adjudication of ranked candidates.
3. Calculate precision, recall, F1, and a confusion matrix if registered truth becomes available.
4. Reuse radius-1/radius-2/radius-3 outputs as a future registration-improvement baseline.

Appendix A. Deliverable Manifest

Part	Artifact group	Contents
1	threshold_optimizer/	Three masks, slice comparisons, and selection memo.
1	unitcell_mask.npy; unitcell_skeleton.npy	Selected segmentation and skeleton arrays.
1	nde_report/	NDE report and two 3D renders.
1	registered_strut_screen/	Registered strut screen CSV and JSON.
2	metadata/; graph_comparison/	Dataset inventory and graph artifacts.
2	refined_registration_20260728/	Authoritative sensitivity readout and radius runs.
2	tube_r2/strut_candidates.csv	All 18,468 records for retained setting.
2	registered_screen_20260728/	Supplemental overlay, provenance, sensitivity, interactive scene.
2	visual_repair*/	Historical non-FEA candidate-repair scenarios.

Appendix B. Reproducibility

Part 1 source: data/unitcell/unitcell.npy. Part 2 CT: data/missing_struts/tif_stacks/210127_Brian_Tran_strut_lattices_0point5dash1 1 Slices.tif. Part 2 design: data/missing_struts/registered_jsons/210127_Brian_Tran_strut_lattices_0point5dash1 1 Slices.json. Authoritative output: output/part2/refined_registration_20260728/tube_r2/. Complete artifact listing: output/DELIVERABLES.md.